EXP 7

Basic R Programs

A. Find the Factors of a Number

Aim:

To write an R program to find all factors of a given number.

Procedure:

1. Read a positive integer.
2. Check the numbers from 1 to the given number.
3. Print the numbers that divide it without a remainder.

Code:

```
n <- 12

cat("Factors of", n, "are:\n")

for (i in 1:n) {
  if (n %% i == 0) {
    cat(i, " ")
  }
}
```

Output:

```
Factors of 12 are:
1  2  3  4  6  12
```

B. Generate 10 Random Integers Between -50 and 50

Aim:

To generate and display 10 random integers between -50 and 50.

Procedure:

1. Use the `sample()` function.
2. Generate 10 integers from the range -50 to 50.
3. Display the generated numbers.

Code:

```
random_numbers <- sample(-50:50, 10)

print(random_numbers)
```

Output:

```
 [1] -12 -50  -2  26  48 -28  10 -11  -6   4
```

Result:

Thus, 10 random integers between -50 and 50 were generated successfully.

Note: The output changes each time the program runs.

C. FizzBuzz from 1 to 100

Aim:

To print numbers from 1 to 100 using FizzBuzz logic.

Procedure:

1. Use a loop from 1 to 100.
2. Print `FizzBuzz` for multiples of both 3 and 5.
3. Print `Fizz` for multiples of 3.
4. Print `Buzz` for multiples of 5.
5. Print the number otherwise.

Code:

```
for (i in 1:100) {
  if (i %% 3 == 0 && i %% 5 == 0) {
    print("FizzBuzz")
  } else if (i %% 3 == 0) {
    print("Fizz")
  } else if (i %% 5 == 0) {
    print("Buzz")
  } else {
    print(i)
  }
}
```

OUTPUT:

```
[1] 1
[1] 2
[1] "Fizz"
[1] 4
[1] "Buzz"
[1] "Fizz"
[1] 7
[1] 8
[1] "Fizz"
[1] "Buzz"
[1] 11
[1] "Fizz"
[1] 13
[1] 14
[1] "FizzBuzz"
[1] 16
[1] 17
[1] "Fizz"
[1] 19
[1] "Buzz"
[1] "Fizz"
[1] 22
[1] 23
[1] "Fizz"
[1] "Buzz"
[1] 26
[1] "Fizz"
[1] 28
```

```
[1] 29
[1] "FizzBuzz"
[1] 31
[1] 32
[1] "Fizz"
[1] 34
[1] "Buzz"
[1] "Fizz"
[1] 37
[1] 38
[1] "Fizz"
[1] "Buzz"
[1] 41
[1] "Fizz"
[1] 43
[1] 44
[1] "FizzBuzz"
[1] 46
[1] 47
[1] "Fizz"
[1] 49
[1] "Buzz"
```

```
[1] "Fizz"
[1] 52
[1] 53
[1] "Fizz"
[1] "Buzz"
[1] 56
[1] "Fizz"
[1] 58
[1] 59
[1] "FizzBuzz"
[1] 61
[1] 62
[1] "Fizz"
[1] 64
[1] "Buzz"
[1] "Fizz"
[1] 67
[1] 68
[1] "Fizz"
[1] "Buzz"
[1] 71
[1] "Fizz"
[1] 73
[1] 74
[1] "FizzBuzz"
[1] 76
```

```
[1] 77
[1] "Fizz"
[1] 79
[1] "Buzz"
[1] "Fizz"
[1] 82
[1] 83
[1] "Fizz"
[1] "Buzz"
[1] 86
[1] "Fizz"
[1] 88
[1] 89
[1] "FizzBuzz"
[1] 91
[1] 92
[1] "Fizz"
[1] 94
[1] "Buzz"
[1] "Fizz"
[1] 97
[1] 98
[1] "Fizz"
[1] "Buzz"
```